RECOMMENDATION SYSTEM: API CHEAT SHEET

Tools, Free APIs & Datasets to Implement Every Recommendation Feature

Internal Team Reference Guide | ProofPass Hackathon Project

Note for Team Members: We do NOT need to collect massive datasets or train our own machine learning models from scratch. Each recommendation feature below uses lightweight free APIs, open datasets, and mathematical formulas that can be built in hours.

1. Course & Practice Recommendations (/recommendations)

YouTube Data API v3

[100% Free (10,000 units/day)]

Feature Target: Course & Video Tutorial Recommendations

How it Helps: Automatically searches top-rated, modern video courses for any missing skill and returns video thumbnails and channel names.

Implementation: *Send GET request: https://www.googleapis.com/youtube/v3/search?q={skill}+tutorial+project&type=video*

GitHub Public Search API

[100% Free (5,000 requests/hr)]

Feature Target: Real-Work Practice & Open-Source Tasks

How it Helps: Fetches live beginner-friendly bugs from active GitHub repos so students write real code to verify their skills.

Implementation: *Send GET request: https://api.github.com/search/issues?q=label:good-first-issue+language:{skill}*

Kaggle Free Tech Courses Dataset

[100% Free (Offline 3MB CSV)]

Feature Target: Coursera & Udemy Course Recommendations

How it Helps: Provides an instant offline database of 30,000+ real tech courses with ratings and URLs. Guarantees 0-cost and 0-latency.

Implementation: *Load CSV into SQLite (proofpass.db). Query: SELECT title, url FROM courses WHERE skill LIKE '%{skill}%' LIMIT 3;*

2. Skill-Gap Roadmaps (/learnings)

Roadmap.sh Open-Source Repository

[100% Free & Open Source]

Feature Target: Standard Step-by-Step Career Roadmaps

How it Helps: The 280,000-star project provides structured JSON and markdown roadmaps for every tech role (Frontend, Backend, DevOps, etc.).

Implementation: *Fetch or copy raw JSON files directly from GitHub: github.com/kamranahmedse/developer-roadmap*

Google Gemini 1.5 Flash API

[100% Free Tier (15 requests/min)]

Feature Target: Dynamic Custom Milestones & Study Tasks

How it Helps: Generates a customized 4-step checklist to upgrade a specific skill from Claimed to Verified in 400 milliseconds.

Implementation: *Prompt Gemini with response_format={'type': 'json_object'}: 'Create 4 steps to verify [skill] with practical tasks.'*

3. Best-Matching Job Roles (Applicant Dashboard)

Weighted Math Matching Engine [Free (Pure Code, No External API)]

Feature Target: Deterministic Job Match Percentage

How it Helps: Calculates transparent percentage scores (0 to 100%) based on job requirements (Must/Should/Nice) vs candidate states (V/S/C/X).

Implementation: *10 lines of Python or JS math: Sum(Skill_Score * Importance_Weight) / Max_Possible_Weight * 100. Runs in 0.001 ms.*

Sentence-Transformers / Gemini Embeddings [100% Free]

Feature Target: Semantic Skill Matching (Synonym Handling)

How it Helps: Matches skills even if words differ (e.g. recognizes that 'FastAPI' matches 'Python REST APIs' with 94% semantic similarity).

Implementation: *Use Python sentence-transformers ('all-MiniLM-L6-v2') or free Gemini embeddings to calculate cosine similarity.*

4. Nearest Companies & Internships (Offline / Online)

HTML5 Geolocation + Haversine Math [Free (Native Browser + 5 Lines Math)]

Feature Target: Automatic Distance Calculation (KM)

How it Helps: Browser detects the candidate's latitude/longitude; a 5-line math formula calculates exact distance to local company offices.

Implementation: *navigator.geolocation.getCurrentPosition() -> compute distance to company lat/long using standard Haversine formula.*

JSearch API (RapidAPI) [Free Tier (500 requests/month)]

Feature Target: Live Job & Internship Postings

How it Helps: Scrapes live job openings directly from LinkedIn, Indeed, and Glassdoor with exact office addresses, logos, and salaries.

Implementation: *Call RapidAPI: GET /search?query='React Intern in Pune'&num_pages=1 -> parses live jobs and office coordinates.*

Mapbox GL API [Free Tier (50,000 map loads/month)]

Feature Target: Interactive Dark-Mode Radar Map UI

How it Helps: Renders glowing burnt-orange pins for high-matching nearby companies on an Apple-Maps quality dark theme.

Implementation: *React component using react-map-gl or mapbox-gl. Display pins for jobs with >80% match within 15 km.*